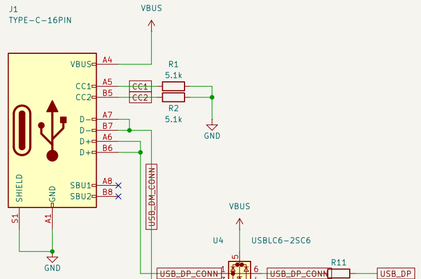
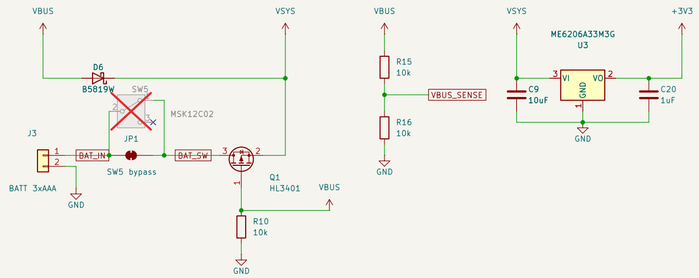
INTENT 2026 hardware badge — generated by 03-kicad/scripts/gen_sch.py from design.py (edit design.py, not this file).

USB-C (USB 2.0 full speed, device)



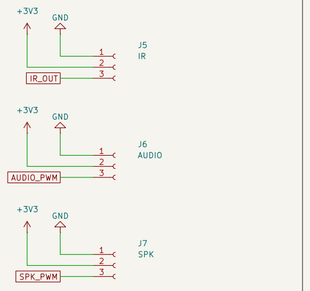
5.1k on CC1/CC2 = UFP (device). SBU pins unused.
USBLC6-2 ESD array in line, 27R series resistors
per the RP2040 hardware design guide.

POWER USB or 3xAAA -> VSYS -> 3V3 LDO



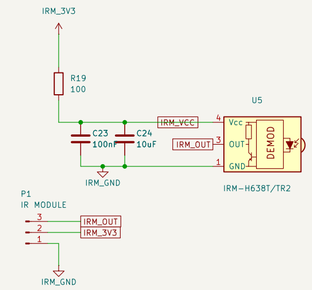
VSYS = 4.65 V on USB (after D6), 3.0–4.8 V on battery. With both present VBUS holds Q1 off, D6 blocks back-feed.
SW5 is DNP (the switch is on the battery box); JP1 bridges it. Cut JP1 only if SW5 is fitted.
RGB LEDs run from VSYS, everything else from 3V3 (ME6206, 300 mA).
R15/R16: VBUS_SENSE = 2.5 V on USB, 0 V on battery (GPIO29 / ADC3) so firmware can dim and sleep.

EXPANSION PORTS 3 x 1x3 socket



Same pinout on all three: 1 = GND (square pad), 2 = 3V3, 3 = signal.
Power in the middle: a module plugged in backwards swaps GND
and signal, never 3V3 onto a GPIO.
IR -> GPIO22, AUDIO -> GPIO11, SPK -> GPIO25 (also J2 pin 15).

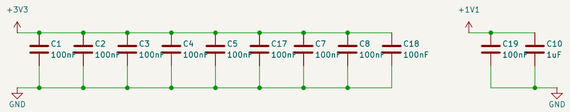
IR MODULE breakaway board, plugs into J5

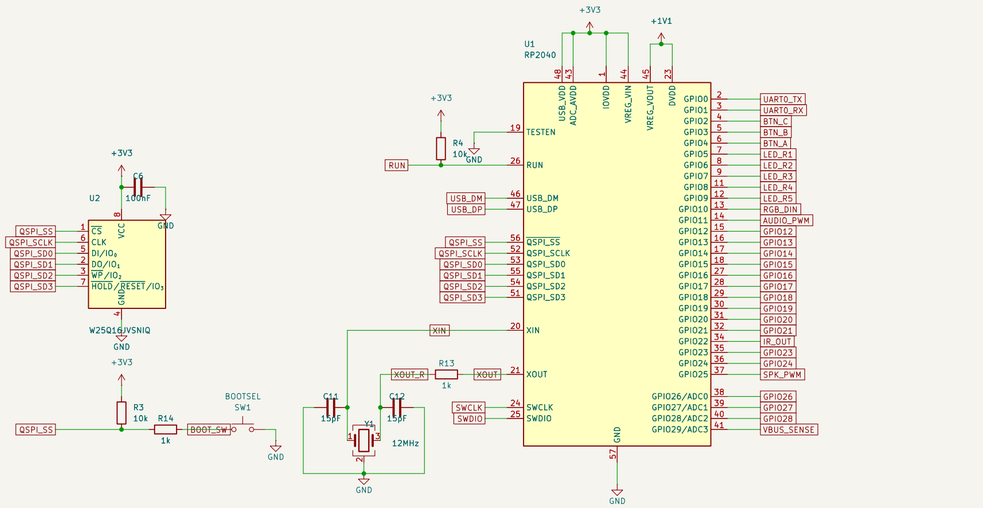


Everlight IRM-H638T, 38 kHz. Open-collector out, active low, internal pull-up.
R19 + C24 = datasheet RC filter. Pinout 1,2 = GND, 3 = OUT, 4 = VCC (verified).
P1: 1 = GND, 2 = 3V3, 3 = OUT, mates with J5 (IR slot). IRM_* nets live only on this board;
no copper crosses the tab. Snap off at the mouse bites, plug in standing up (lens looks sideways).

RP2040 dual Cortex-M0+, 2 MB QSPI flash, 12 MHz crystal

Decoupling: one 100 nF per RP2040 supply pin (6x IOVDD, USB_VDD, ADC_AVDD, VREG_VIN); 1 uF + 100 nF on the 1V1 core rail.

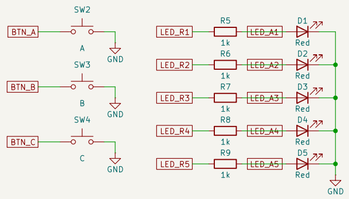




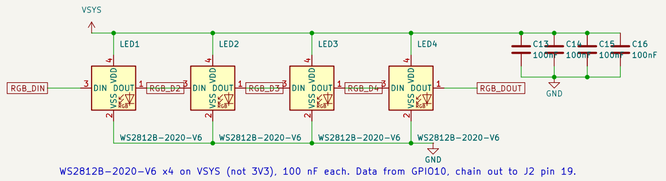
BOOTSEL: hold SW1 while plugging USB in -> UF2 bootloader. R14 protects the QSPI_SS driver.
Crystal 12 MHz: CL = 20 pF part with 15 pF loads (Pico values, ~ +50 ppm, fine for USB). R13 limits drive.
TESTEN tied to GND. RUN has a 10k pull-up and is on J2 for an external reset / debugger.

GPIO: 0/1 UART0 -> TP1/TP2; 4/3/2 = keys A/B/C;
5-9 red LEDs; 10 WS2812 data; 11 AUDIO slot (PWM);
22 IR slot; 25 SPK slot (also J2.15); 29 VBUS sense (ADC3);
12-21, 23/24, 26-28 free on J2 (SPI1, I2C0, UART1, ADC0-2).

USER INTERFACE keys, credit LEDs, RGB shelf

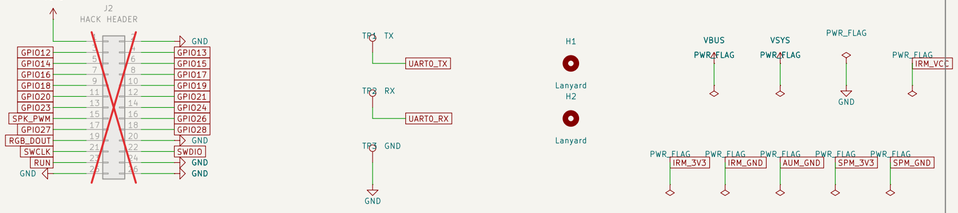


Keys A/B/C: active low, internal pull-ups, debounce in firmware.
Credit display: 5 red LEDs, 1k series -> ~1.5 mA each (dim by design).



WS2812B-2020-V6 x4 on VSYS (not 3V3), 100 nF each. Data from GPIO10, chain out to J2 pin 19.
DIN is driven at 3.3 V against VIH = 0.7 x VDD: marginal on paper, works in practice.

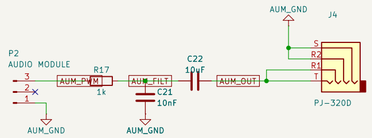
HACK HEADER J2 (DNP), test points, lanyard holes



2x13, 2.54 mm, not fitted — players solder their own pins. 3V3 logic only, no 5 V on the header.
Pin 1 = top-left square pad. GND on both end rows: a reversed ribbon shorts 3V3 to GND at worst (LDO limits).
15 free GPIO + SPK_PWM + SWD + RUN + RGB chain out. TP1-TP3: UART0 console 115200 8N1. H1/H2: 4.0 mm plated lanyard holes.

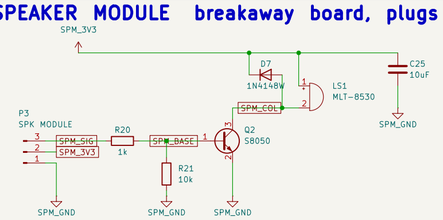
PWR_FLAGs: nets fed only through passive pins
(row 2: breakaway-module rails and grounds).

AUDIO MODULE breakaway board, plugs into J6



GPIO11 PWM (>= 100 kHz carrier) -> 1k / 10n low-pass (fc ~ 16 kHz) -> 10 uF DC block -> tip + ring 1 (mono). Sleeve + ring 2 = GND.
Level into 32 ohm ~ 0.1 Vpp. PJ-320D (C431535) SMD pads: 1 = S, 2 = R2, 3 = R1, 4 = T. Detect-switch pads unused.
P2: 1 = GND, 2 = n.c. (no 3V3 needed), 3 = PWM, mates with J6 (AUDIO slot). Snap off; the jack then points sideways.

SPEAKER MODULE breakaway board, plugs into J7



MLT-8530 magnetic transducer: 2.7 kHz resonance, 16 ohm,
<= 95 mA at 2.5-4.5 V, so it runs from the 3V3 LDO.
S8050 low-side switch, 1k base; 10k keeps it off while the
GPIO floats at reset. 1N4148W flyback, 10 uF local bulk.
Drive: 2.7 kHz square wave (50 % PWM) on GPIO25.
GPIO25 is also J2 pin 15 (shared).
P3: 1 = GND, 2 = 3V3, 3 = SIG, mates with J7 (SPK slot).
Footprint wires the two dummy pads crosswise,
so a 180 deg placement only reverses the coil.
(BOM-B2 add-on 7.)

Generated by scripts/gen_sch.py — edit design.py, not this file
RP2040 badge, 56x100 mm vending-machine outline + 3 breakaway modules (IR, audio, speaker)
INTENT 2026 — by Palo Alto Networks

Sheet: /	
File: intent-2026-badge.kicad_sch	
Title: INTENT 2026 Hardware Badge	
Size: A2	Date: 2026-10-11
KiCad E.D.A. 10.0.3	Rev: 0.5
	Id: 1/1